NexusMart: Data Quality Report

What was wrong with the raw data and how it was fixed. Every fix is implemented in Python (scripts/01_clean_lookups.py, 02_clean_orders.py, 03_clean_payments_shipping_returns.py); nothing was edited by hand in Excel, Power Query or the database. Counts below are written by the scripts to data/logs and loaded into the table dw.cleaning_log. Records that cannot be repaired go to data/rejects with a reason code, and are also stored in dw.reject_log.

1. Raw data profile (before cleaning)

File	Rows	Exact duplicate rows	Main problems found
customers	45,000	0	2,233 missing city, 4,600 missing birth year; 7 spellings of gender; mixed phone and date formats; about 2,360 duplicate accounts (same person, different ID)
products	3,200	0	38 spellings of 12 categories; prices as text (Rs. 2,900); price and weight outliers; 170 missing cost price
sellers	420	0	22 missing tier, ratings above 5, mixed date formats
locations	650	0	Province spelled many ways (Panjab, KPK, Sind); 25 missing province; coordinates outside Pakistan
orders	163,200	3,200	Status in about 20 spellings, channel in 12; 3 date formats and 1,605 impossible dates; 6,516 missing ship city; order_total outliers
order_items	315,143	0	Quantity as words (one, 2.0), negative or 500; prices as text, 0 or 9,999,999; 31,567 missing discounts
payments	168,111	3,296	22 spellings of payment method; negative and 9,999,999 amounts; 5,017 missing dates
shipping	151,228	0	22 spellings of courier; 5,943 missing shipped date, 18,841 missing delivered date; delivery before shipping; fee outliers and 9,145 missing
returns	16,048	0	Status spellings (Declined = Rejected); 2,390 missing reason; impossible dates; refunds above price paid

Broken references found: orders to customers 330; order items to products 932; returns to order items 139; all other links were valid.

2. Cleaning decisions and before/after counts

Each row is one rule applied by the code. 'Rows before/after' show row counts where rows changed; otherwise the number of values changed or flagged is shown.

Lookup tables

Table	Rule applied	Rows before	Rows after	Values changed	Note
locations	province spellings standardised			600	-> standard province names
locations	missing province set to 'Unknown'			25	(cannot be repaired from other data)
locations	coordinates outside Pakistan set to blank			10	(outliers)
locations	added 'Unknown' member (-1)	650	651	-1	
sellers	missing tier set to 'Unknown'			22	
sellers	ratings outside 0-5 set to blank			11	(outliers)
sellers	added 'Unknown' member	420	421	-1	
products	category names standardised			3200	-> 12 categories
products	missing brand set to 'Unknown'			392	
products	missing sub_category set to 'Unknown'			485	
products	impossible list_price (<=0 or >1,000,000) set to blank			13	(outliers)
products	impossible weight (<=0 or >100 kg) set to blank			17	(outliers)
products	missing cost_price estimated = list_price x category median ratio			170	(assumption)
products	missing list_price estimated = cost_price / category median ratio			13	(assumption)
products	flagged cost_price > list_price (kept, flag column)			67	
products	added 'Unknown' member	3200	3201	-1	
customers	phones standardised to 03XXXXXXXXX			41401	(of 41401 given; the rest were invalid)
customers	implausible birth_year (<1930 or >2010) set to blank			142	(outliers)
customers	missing/invalid city_id mapped to Unknown (-1)			2233	

Table	Rule applied	Rows before	Rows after	Values changed	Note
customers	duplicate customers merged under one master ID	45000	42640	2360	(2360 duplicate IDs saved in customer_id_map.csv)

Orders and order items

Table	Rule applied	Rows before	Rows after	Values changed	Note
orders	exact duplicate rows removed	163200	160000	3200	
orders	order_status standardised			160000	-> Delivered / Cancelled / Returned / In Transit
orders	channel standardised			160000	-> App / Web / Mobile Web
orders	customer_id not found -> 'C_UNKNOWN' (broken reference)			321	
orders	impossible/unreadable order_date found			1605	(e.g. 2024-13-05, 00/00/2024, years 2026-2027)
orders	order_date repaired = shipped_date - 1 day			1457	
orders	order_date repaired from neighbouring order numbers			148	
orders	missing ship city filled with customer's city (assumption)			6058	
orders	ship city still unknown -> 'Unknown' (-1)			330	
order_items	quantity words/decimals converted to whole numbers (one, 2.0 ...)			5767	
order_items	negative quantity -> absolute value (sign error)			307	
order_items	product_id not found -> 'P_UNKNOWN' (broken reference)			932	
order_items	lines whose order does not exist -> rejects file	315143	315143		
order_items	negative unit_price -> absolute value			316	
order_items	unit_price 0/9999999/blank/scale error (outside 0.5x-2x list price) -> list_price x 1.026 (assumption)			1240	
order_items	quantity outliers (500, 0, blank) rebuilt from order_total (single-line orders)			115	
order_items	quantity outliers in multi-line orders set to 1 (most common value)			526	(assumption)
order_items	blank/impossible discount_amount -> gross x discount % of the order			32447	
orders	order_total negative/9999999/blank -> rebuilt from its lines			488	

Payments, shipping and returns

Table	Rule applied	Rows before	Rows after	Values changed	Note
payments	exact duplicate rows removed	168111	164815	3296	
payments	payment_method (22 spellings) and payment_status standardised			164815	-> 5 methods, 4 statuses
payments	negative amount -> absolute value			663	
payments	amount 9,999,999 placeholder set to blank			162	(outliers)
payments	payment_date missing/impossible left blank			5994	
payments	payments for unknown orders -> rejects file	164815	164815		
payments	orders paid in more than one part (main method = largest part)			4815	
shipping	exact duplicate rows removed	151228	151228		
shipping	courier (22 spellings) and delivery_status standardised			151228	-> 6 couriers
shipping	missing/impossible shipped_date = order_date + 1 day			6965	
shipping	delivered_date before shipped_date -> dates swapped			1335	
shipping	delivery time over 30 days (implausible) -> delivered_date blanked			278	(outliers)
shipping	fee blank/0/over 10,000 -> that courier's median fee (assumption)			9440	(negatives made positive first)
shipping	shipments for unknown orders -> rejects file	151228	151228		
returns	exact duplicate rows removed	16048	16048		

Table	Rule applied	Rows before	Rows after	Values changed	Note
returns	return_status standardised (Declined = Rejected)			16048	-> Refunded / Approved / Rejected
returns	return_reason standardised (33 spellings -> 9 reasons + Unknown)			16048	
returns	Rejected returns marked not valid (kept in table, ignored in metrics)			1322	
returns	returns for unknown order lines -> rejects file	16048	15909	139	
returns	return_date impossible/before order -> order_date + 10 days (median)			256	
returns	refund_amount above the line's paid value -> capped at line value			264	(outliers)
returns	valid return with refund 0/blank -> full line value (assumption)			0	

3. Duplicates, categories, merges and rejects

Exact duplicates: removed 3,200 orders (163,200 to 160,000) and 3,296 payments (168,111 to 164,815); counts logged. **Duplicate customers:** merged 2,360 accounts into 42,640 people. A match needs the same name AND the same e-mail, or the same name AND the same phone; name alone is never used. A mapping table (customer_id_map.csv) keeps every old ID linked to its master ID, so orders still join. **Categories:** 38 spellings standardised into the 12 official categories. **Rejected records:** 139 returns pointing to order lines that do not exist are stored in data/rejects/returns_rejects.csv (reason: order_item_id not in order_items) and in dw.reject_log. **Broken references that were kept:** 321 orders map to the Unknown customer, 932 lines to the Unknown product, 330 orders to the Unknown location (every dimension has an Unknown member).

4. Key assumptions

- Missing ship city: filled with the customer's own city (88.9% of known orders ship to it); orders still unknown go to the Unknown location. Flag column: ship_city_imputed.

- Impossible order dates (1,605): repaired as shipped date minus 1 day (couriers ship about 1 day after the order, checked on valid rows), otherwise copied from the neighbouring order number (order numbers grow with time). All dates now fall between 1 Jan 2024 and 30 Jun 2025.

- Bad unit prices (0, 9,999,999, outside 0.5x to 2x list price): replaced with list price x 1.026 (median price/list ratio on good rows). Missing cost price: list price x category median cost ratio.

- Quantity outliers (500, 0): rebuilt from the order total for single-line orders (115), otherwise set to 1, the most common value (526). Negative quantities and prices made positive (sign errors).

- Blank or impossible discounts: recalculated as gross amount x the order's discount-code percentage. Order totals are rebuilt from lines when invalid; 99.15% of order totals equal the sum of their lines within 1 PKR after cleaning.

- Delivered before shipped (1,335 shipments, always exactly 2 days off): treated as swapped dates. Deliveries over 30 days: delivered date blanked. Missing/invalid courier fees: that courier's median fee. Fees are split over order lines by value, so they add back to the real total.

- Returns: status Rejected (including Declined) is kept in the data but is not a valid return. Refund amount is capped at what the customer paid for the line. Impossible return dates become order date + 10 days (median).

5. Checks run by the pipeline (quality_checks task)

Fact row count equals the cleaned order lines (315,143); no duplicate order_item_id (grain respected); net amount equals the source; shipping fees are not double counted; share of lines on the Unknown member below 5% per dimension (actual: customer 0.20%, product 0.30%, seller 0.00%, location 0.21%). The pipeline stops with an error if a check fails. Running it twice gives identical row counts (idempotent).

6. Known limitations

Products missing from the catalogue (0.3% of lines) have no cost price, so margin on them is not meaningful. One order line has a price near zero for the same reason. 25 cities have an unknown province (6.4% of NMV). Fill rules above are assumptions, listed so a different choice can be compared.